

GSIP Simulation Report

Run: SpatialPack diffusion (in-process, no Temporal)
Mode: domain_pack | Status: completed
Pack: spatial-pack | Created: 2026-08-27T10:01:55

Your Question

Reduce pollution concentration while keeping coverage of the monitored area

What This Simulation Did

The engine evaluated 8 scenarios. Each scenario applies a different combination of policy levers to a deterministic model, scores the outcomes, and ranks the best options. The LLM is used only to set up the problem and write this report - not once per scenario.

Answer / Recommended Solution

In-process execution of spatial-pack without Temporal. 8 scenarios were simulated by the domain-pack simulate() method. This is not a Temporal-orchestrated run.

Executive Summary

In-process execution of spatial-pack without Temporal. 8 scenarios were simulated by the domain-pack simulate() method. This is not a Temporal-orchestrated run.

Results Summary

Scenarios simulated: 8 / 8
Best judge score: 0.695
Mean judge score: 0.643
Finalists promoted to higher fidelity: 5

Best Scenario Detail

Scenario ID: spatial-006
Judge score: 0.695 (acceptable)
Optimal lever settings:
- sources: [{ 'x': 24, 'y': 20, 'intensity': 0.7235456444608611, 'radius': 2.802862157376354}, { 'x': 27, 'y': 9, 'intensity': 0.4529194998038859, 'radius': 1.4015379083224386}, { 'x': 14, 'y': 12, 'intensity': 1.1502530614759756, 'radius': 2.8122686777911214}]
Outcome metrics:
- coverage_ratio: 0.673
- max_concentration: 0.610
- mean_concentration: 0.072
- safe_area_ratio: 0.993
- warning_area_ratio: 0.007
- critical_area_ratio: 0.000
- threshold_violations: 0.000

Top Ranked Alternatives

Rank	Score	Key levers / metrics
1	0.695	coverage_ratio=0.673, max_concentration=0.610, mean_concentration=0.072
2	0.667	coverage_ratio=0.589, max_concentration=0.647, mean_concentration=0.049
3	0.666	coverage_ratio=0.584, max_concentration=0.321, mean_concentration=0.037
4	0.665	coverage_ratio=0.580, max_concentration=0.435, mean_concentration=0.040
5	0.664	coverage_ratio=0.590, max_concentration=0.764, mean_concentration=0.091
6	0.636	coverage_ratio=0.487, max_concentration=0.522, mean_concentration=0.036

7	0.587	coverage_ratio=0.325, max_concentration=0.463, mean_concentration=0.021
8	0.562	coverage_ratio=0.290, max_concentration=1.083, mean_concentration=0.064

Important Notes

Results depend on the registered domain pack fidelity and evidence base. Review benchmarks and assumptions before acting on recommendations.

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